User Manual

for

UniPool

Version 1.0

Prepared by

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Contents

CONTENTS II

REVISIONS III

1 INTRODUCTION 1

1.1 TEST STRATEGY 1
1.2 TESTING TIMELINE 1
1.3 TESTERS 1
1.4 COVERAGE CRITERIA 1
1.5 TOOLS USED FOR TESTING 1

2 UNIT TESTING 3

2.1 AUTHENTICATION 3
2.2 PROFILE MANAGEMENT 14
2.3 RIDE HANDLING 18
2.4 CHAT/MESSAGING 27

3 INTEGRATION TESTING 32
3.1 AUTHENTICATION 32
3.2 CREATE RIDE POST 38
3.3 FIND AND FILTER RIDES 42
3.4 EDIT PROFILE 47
3.5 CHAT SYSTEM 50
3.6 MY ACTIVITY 55
4 SYSTEM TESTING 65
5 CONCLUSION 69

APPENDIX A – GROUP LOG 71

Revisions

Version	Primary Author(s)	Description of Version	Date Completed
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INTRODUCING UNIPOOL

Purpose

UniPool is a dedicated campus ride-sharing mobile application built exclusively for the IIT Kanpur community. It directly addresses one of the most persistent and frustrating daily challenges faced by students: the difficulty of coordinating affordable, safe, and timely transportation between the campus and key city destinations such as Kanpur Central Railway Station, Lucknow Airport, Z Square Mall, SPM Hospital, and Naveen Market.

Traditionally, students relied on fragmented WhatsApp groups, notice-board announcements, or informal word-of-mouth to find fellow travellers willing to share an auto-rickshaw or cab. This approach is slow, unreliable, and often means students end up booking expensive solo rides — particularly during exam seasons, festival breaks, or early-morning train departures.

UniPool eliminates these barriers by introducing a structured, real-time, and transparent platform. It provides two dedicated modes of operation:

- **Leader mode — Any student can post a ride request by specifying the source, destination, and date. The ride becomes immediately visible to all registered users.**
- **Pooler mode — Students who need a ride can browse all open rides, filter by destination, review the ride leader's profile and reputation, and join the group chat to coordinate pickup details.**

By connecting students who are already travelling the same route, UniPool significantly reduces per-person transport costs. It also builds a community of trust by tracking ride history and surfacing reputation scores for each ride leader.

Key Objectives

- Reduce the cost of off-campus travel by enabling ride-sharing among students travelling the same route.
- Provide a single, organised platform for posting and discovering rides — no more scattered group messages.
- Enable real-time coordination through per-ride group chat so participants can confirm pickup points and timing.
- Build community trust through ride completion counters and profile verification.
- Support seamless authentication via college email or Google Sign-In, ensuring only verified IITK community members can participate.

Software Overview

UniPool is developed using the Flutter framework (Dart language) for the frontend and Google Firebase for the entire backend. This architecture delivers a fast, visually polished, and fully real-time experience optimised for Android devices, with a straightforward path to iOS compatibility.

Component	Technology Used
Frontend Framework	Flutter 3.x (Dart)
Authentication	Firebase Authentication (Email/Password + Google Sign-In)
Database	Cloud Firestore (NoSQL, real-time)
File Storage	Firebase Storage (profile photos)
Real-time Chat	Cloud Firestore streams (live message subscriptions)
Build System	Gradle (Android) via Android Studio

Campus Integration

UniPool is built exclusively for IIT Kanpur and is deeply integrated with the campus geography. The location list is curated to reflect the most frequently used pickup and drop-off points — from individual hostels to major city landmarks — making it immediately relevant to every student.

On-campus locations include Hall 1, Hall 2, Hall 3, Hall 4, Hall 12, Hall 13, the Academic Area, the Library, the Main Gate, the Health Centre, and the Shopping Centre. Off-campus destinations include Kalyanpur Metro, SPM Hospital, Vishwavidyalaya, Gurudev Chauraha, Geeta Nagar, Rawatpur, GSVM Medical College, Moti Jheel, Chunniganj, Naveen Market, Bada Chauraha, Nayaganj, Kanpur Central Railway Station, Lucknow Airport, and Z Square Mall.

Intended Audience

UniPool is intended solely for registered members of the IIT Kanpur community, including students, research scholars, and faculty. Every user is verified through their institute email address or Google account, ensuring that all interactions remain secure and trustworthy. The platform is not accessible to individuals outside the IITK community.

INSTALLATION GUIDELINES

Follow the steps below to set up and run UniPool on your local development machine:

Step 1 — Clone the Repository

Open a terminal and run the following command to clone the project source code:

```
git clone <repository_url>
```

Step 2 — Open in Android Studio

Navigate to the cloned project directory and open it using Android Studio. Ensure that the Flutter and Dart plugins are installed and that a valid Flutter SDK path is configured in Android Studio's settings.

Step 3 — Configure Firebase

UniPool requires a valid Firebase project. Place the google-services.json file (obtained from your Firebase console) into the android/app/ directory. Without this file, the app will not build successfully.

Step 4 — Install Dependencies

In the project root directory, run the following command to fetch all required Dart packages:

```
flutter pub get
```

Step 5 — Build and Run

Connect a physical Android device or launch an emulator, then run:

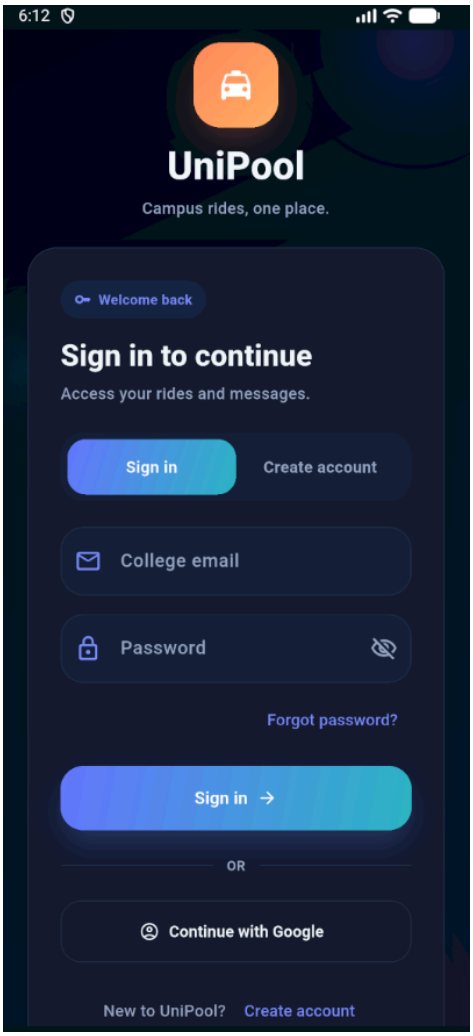
```
flutter run
```

Note: Ensure that the Flutter SDK, Android SDK, and all required platform tools are correctly installed on your system before starting. The minimum required Dart SDK version is 3.10.7.

WELCOME PAGE

Welcome Page

The Welcome Page is the first screen a user sees upon launching UniPool. It serves as the secure gateway to the platform, presenting two clear paths based on whether the user already has an account.



Core Functionalities

- Sign In — Allows existing registered users to authenticate and access all platform features, including browsing rides, creating rides, and messaging.
- Create Account — Guides new users through the registration process using their college email address and a secure password.
- Continue with Google — Enables one-tap sign-in using a Google account, bypassing the manual email and password entry entirely.

- Forgot Password — A quick-access link that sends a password-reset email to the user's registered address.

Redirection and Navigation

If a user successfully authenticates, they are immediately redirected to the Home Screen. If a user logs out from any part of the application, they are returned to this Welcome Page. During the brief moment while Firebase checks authentication status, a loading indicator is shown to prevent a blank screen.

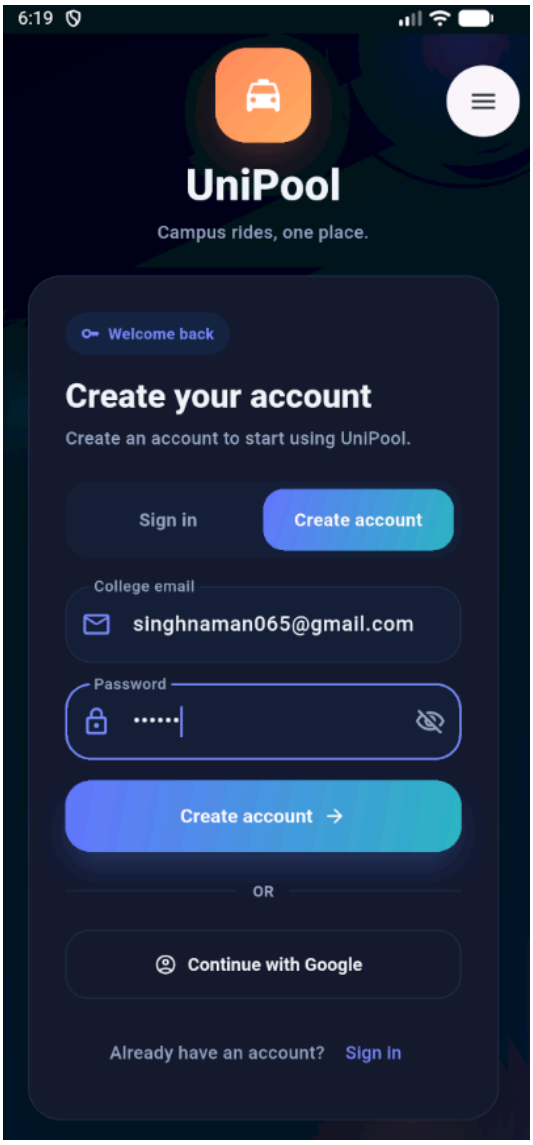
SIGN UP

Sign Up Process

New users can register for UniPool using their email address. The process is intentionally simple and includes an email verification step to ensure all accounts belong to genuine IITK community members.

Registration Steps

1. Tap 'Create account' on the Welcome Page to switch to the registration view.
2. Enter your email address in the 'email' field.
3. Create a secure password of at least 6 characters. Tap the eye icon to toggle password visibility.
4. Tap 'Create account'. UniPool will create your account and dispatch a verification email.
5. Open the verification email in your inbox and click the confirmation link.
6. Return to the app and sign in. Unverified accounts cannot log in.



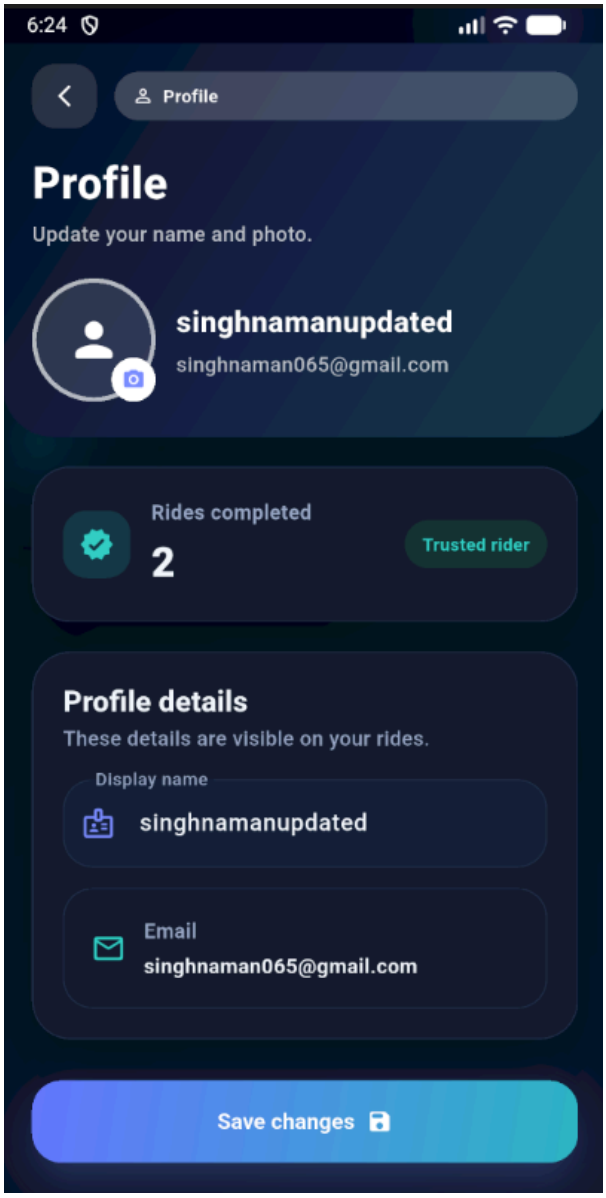
Important Notes

- Your password must be at least 6 characters long.
- If you attempt to sign in before verifying your email, the app will automatically resend a fresh verification link and sign you out.
- Once your account is created, you can also use 'Continue with Google' for faster future sign-ins.

CREATE PROFILE

Creating a New User Profile

After registering and verifying your email, your UniPool profile is automatically initialised with basic information derived from your email address. You can then personalise your profile at any time from the Profile screen. A complete and accurate profile helps other students identify and trust you as a ride leader.



Profile Information

- **Display Name — Enter the name you want other riders to see. By default, this is derived from your email username (e.g., 'john.doe' from john.doe@iitk.ac.in).**
- **Email Address — Your registered email is displayed here for reference. It cannot be edited, as it is the primary authentication identifier.**
- **Profile Photo — Tap the camera icon on the circular avatar to select an image from your device's gallery. The image is automatically uploaded to Firebase Storage.**
- **Rides Completed — This counter is incremented each time a ride you lead is marked as completed. It is visible to other users when they view a ride you have posted.**

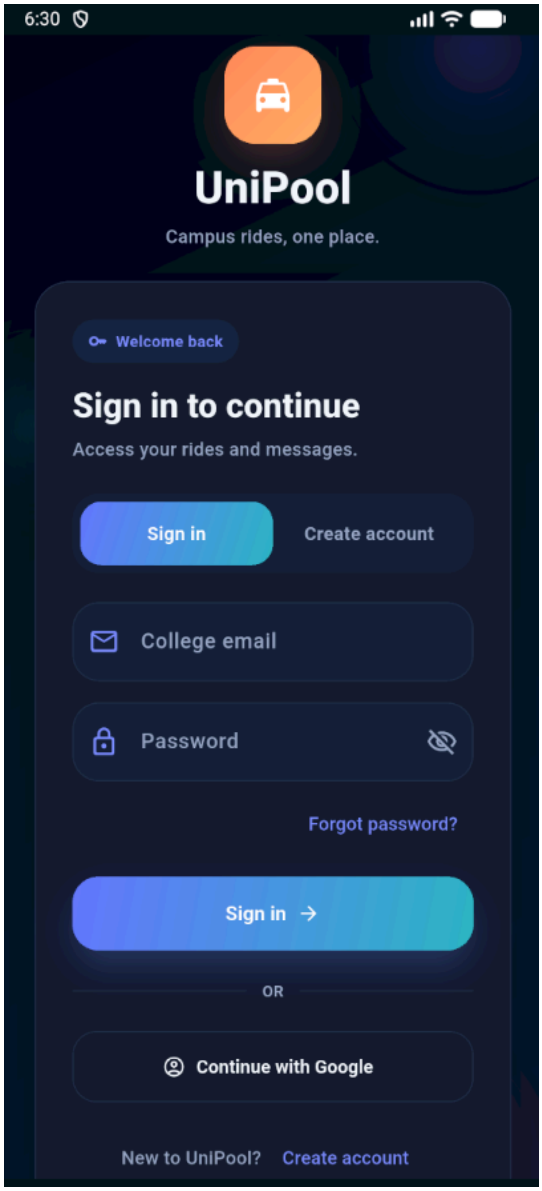
Saving Your Profile

After updating any fields, tap 'Save changes'. A green confirmation message will appear at the bottom of the screen once the changes are successfully saved to the database.

LOGIN PAGE

Login Process

Registered and verified users can securely sign in to UniPool using either their email and password or their Google account. The login process is protected against brute-force attempts and includes clear, actionable error messages.



Email and Password Sign-In

7. Select 'Sign in' on the Welcome Page (it is selected by default).
8. Enter your registered email address.
9. Enter your password. Tap the eye icon to toggle visibility.
10. Tap 'Sign in'. Upon successful authentication, you will be taken directly to the Home Screen.

Google Sign-In

Tap 'Continue with Google' to sign in with a Google account in a single step. UniPool will automatically create your profile if this is your first time signing in with Google. A loading spinner is shown while the authentication completes

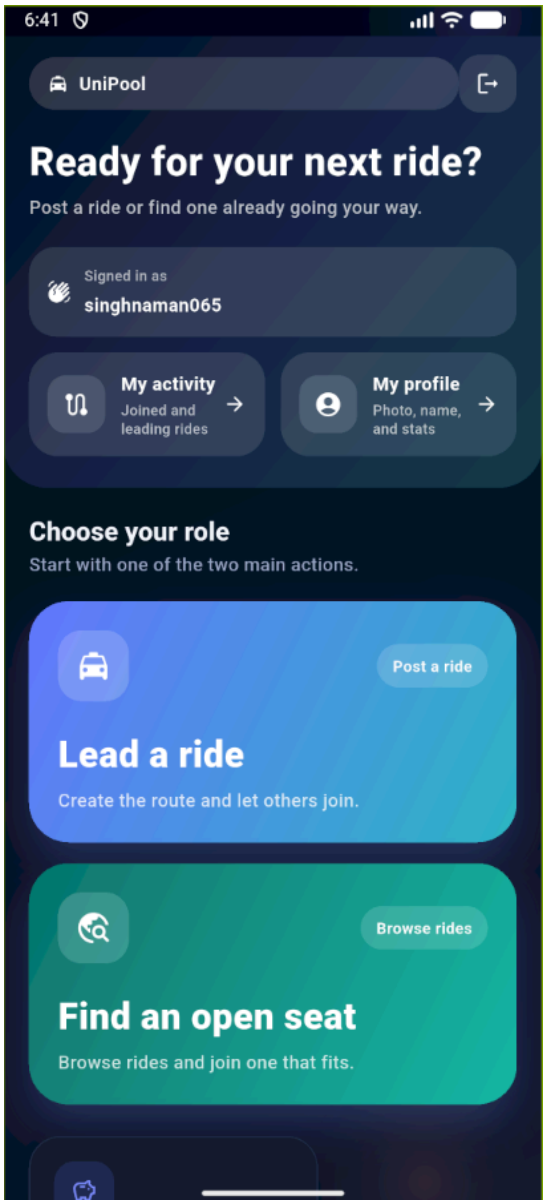
Common Error Messages

Error Message	What It Means
No account found with this email.	The email address has not been registered. Please sign up first.
Incorrect password.	The password entered does not match. Check for typos or use Forgot Password.
Verify your email before signing in.	The account exists but the email address has not yet been confirmed.
Too many attempts. Please wait.	Multiple failed login attempts detected. Wait a few minutes before trying again.

HOME PAGE & DASHBOARD

Home Page Overview

Upon successful login, users land on the Home Screen — the central hub of UniPool. This screen provides a quick overview of who is signed in, direct access to the two primary actions, and links to the user's activity and profile management.



Header Section

- **UniPool brand badge — Displayed in the top-left of the header for immediate brand recognition.**
- **Logout button — A small icon button in the top-right of the header. Tapping it immediately signs the user out and returns them to the Welcome Page.**
- **Signed-in greeting — The header shows a welcome message displaying the currently logged-in user's name, derived from their registered email.**

Quick Action Cards

Two compact cards sit beneath the greeting, providing fast access to secondary screens:

- **My Activity — Navigates to the My Rides screen, where users can view all rides they are leading or have joined.**
- **My Profile — Navigates to the Profile screen for editing display name, profile photo, and viewing ride statistics.**

Primary Action Cards

The main scrollable body of the Home Screen features two large, visually prominent cards:

- **Lead a Ride — Opens the Create Ride screen. This is for students who want to coordinate a ride and invite others to join them.**
- **Find an Open Seat — Opens the Find Ride screen. This is for students looking to join an existing ride that matches their route.**

Benefit Tiles

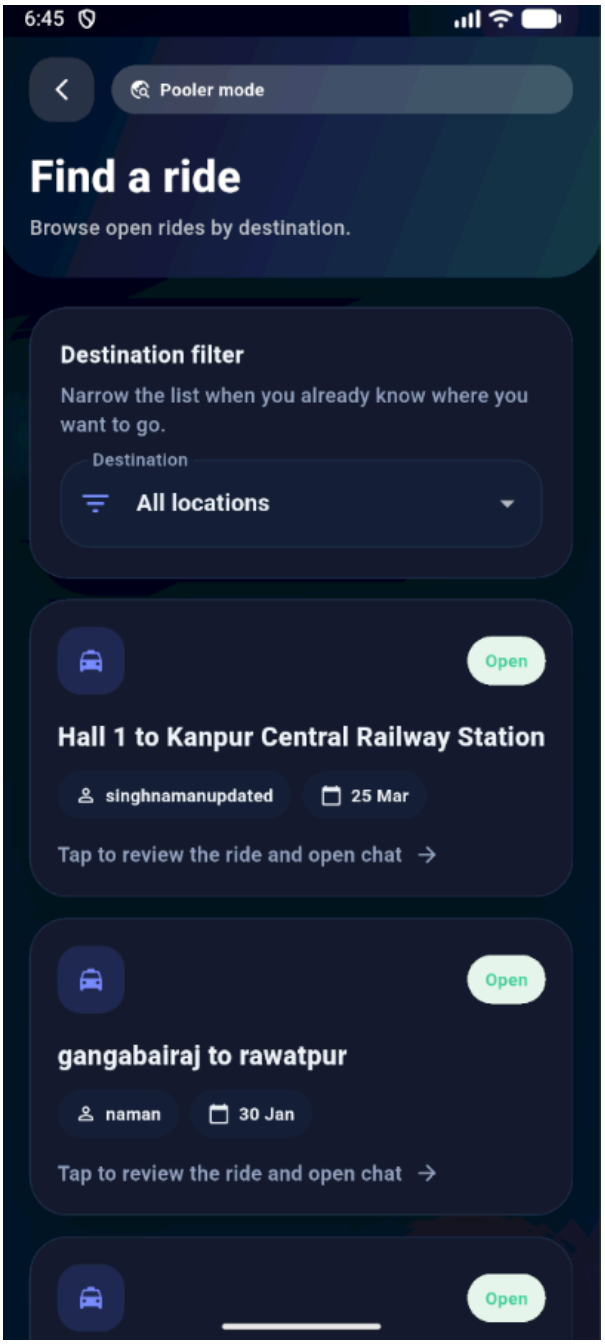
Below the primary action cards, three benefit tiles communicate UniPool's core value proposition at a glance:

- Lower Cost — Split rides instead of booking solo.
- Fast Coordination — Rides and chat stay in one place.
- Fewer Empty Seats — Share trips when routes overlap.

FIND A RIDE (SEARCH & FILTER)

Find a Ride Screen

The Find a Ride screen is the Pooler Mode interface. It displays all rides that are currently open and available for new members to join. Users can apply a destination filter to narrow results and tap any ride card to see full details and enter the group chat.



Destination Filter

A filter card at the top of the screen allows users to narrow the ride list by destination. The filter defaults to 'All locations', showing every open ride. When a specific destination is selected, only rides heading to that location are shown. This is particularly useful during peak travel times when many rides are available simultaneously.

To apply a filter:

11. Tap the 'Destination' dropdown field in the filter card.
12. Select your destination from the full list of campus and city locations.
13. The ride list updates instantly to show only matching rides.
14. To reset, select 'All locations' from the dropdown.

Ride Cards

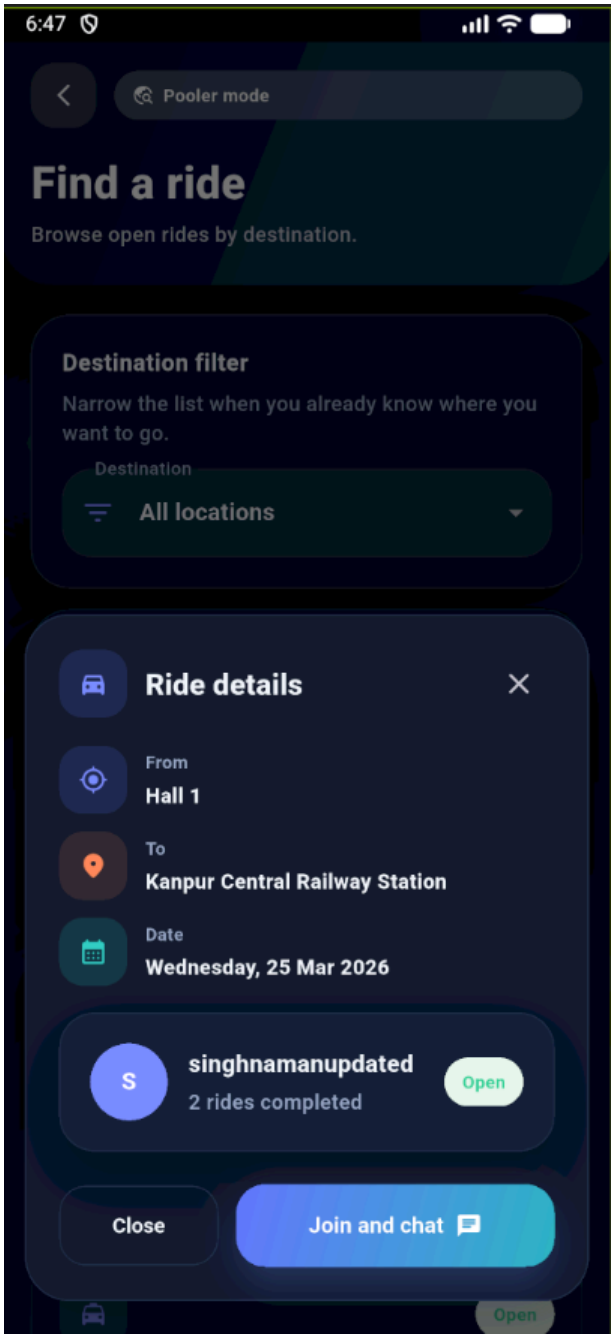
Each ride in the list is displayed as a card showing the following information at a glance:

- Route — Source and destination locations (e.g., 'Hall 5 to Kanpur Central Railway Station').
- Leader — The display name of the student who created the ride.
- Date — The scheduled date for the ride.
- Status Badge — All rides displayed here are 'Open', meaning they accept new participants.

Ride Details Bottom Sheet

Tapping a ride card opens a bottom sheet with the complete ride details. This includes:

- **Full route (From / To / Date) with icons for each field.**
- **Leader profile card showing their avatar, display name, number of rides completed, and open status badge.**
- **Two action buttons: 'Close' to dismiss the sheet and 'Join and chat' to enter the ride's group chat.**



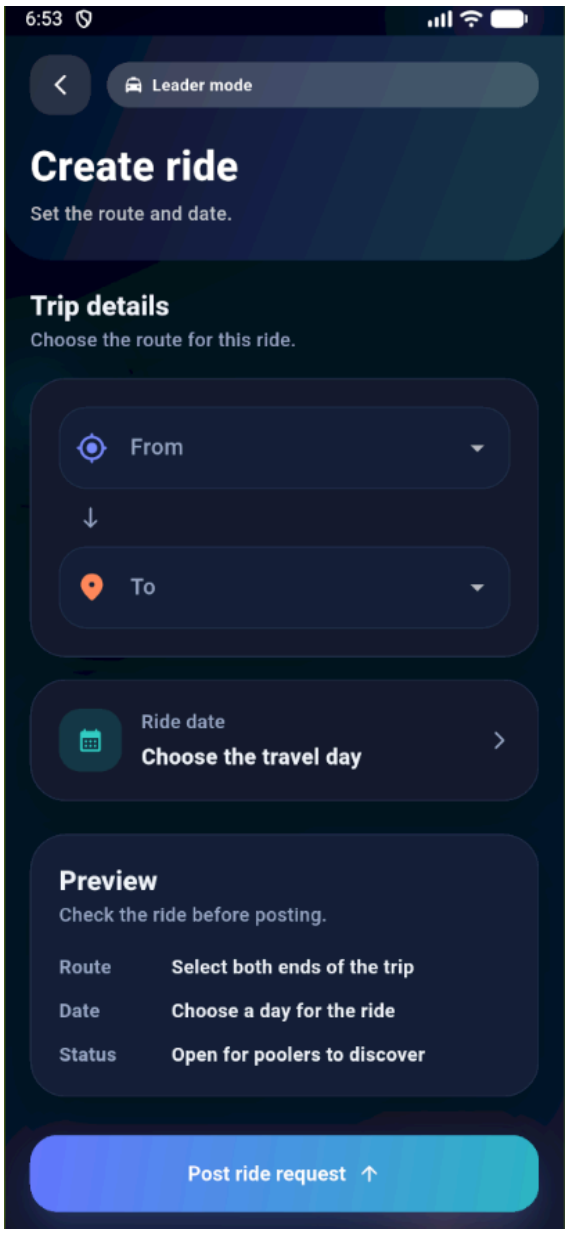
Joining a Ride

Tapping 'Join and chat' on the Ride Details sheet navigates directly to the Ride Chat screen. The first message a user sends automatically adds them to the ride's participant list. There is no explicit 'Join' button — participation is established through chat engagement, keeping the process lightweight and natural.

OFFER A RIDE (CREATE A RIDE)

Create Ride Screen

The Create Ride screen is the Leader Mode interface. It allows any registered user to post a new ride request by specifying the route and travel date. Once posted, the ride is immediately visible to all users browsing the Find a Ride screen.



Completing the Ride Form

The ride creation form consists of three required components:

1. Route Selection

- From (Source) — Select the pickup location from the dropdown list. This defaults to the campus hall or location where you will be departing.
- To (Destination) — Select the drop-off location from the same list. The source and destination must be different; the app will show an error if they match.

2. Date Selection

- Tap the date card to open a calendar date picker.
- Select the date on which the ride will take place. Only the current date and future dates are available.
- The selected date is displayed on the card in a human-readable format (e.g., 'Friday, 28 Mar 2025').

3. Live Preview

As you fill in the form, a 'Preview' card at the bottom of the screen dynamically updates to show the selected route, date, and the ride's initial status ('Open for poolers to discover'). Review this preview before submitting to ensure accuracy.

Posting the Ride

Once all three components are filled, tap 'Post ride request'. UniPool will:

- Create a new ride document in Cloud Firestore with your user ID, display name, route, date, and 'open' status.
- Dismiss the Create Ride screen and return you to the Home Screen.
- Display a green confirmation snackbar: 'Ride posted successfully.'

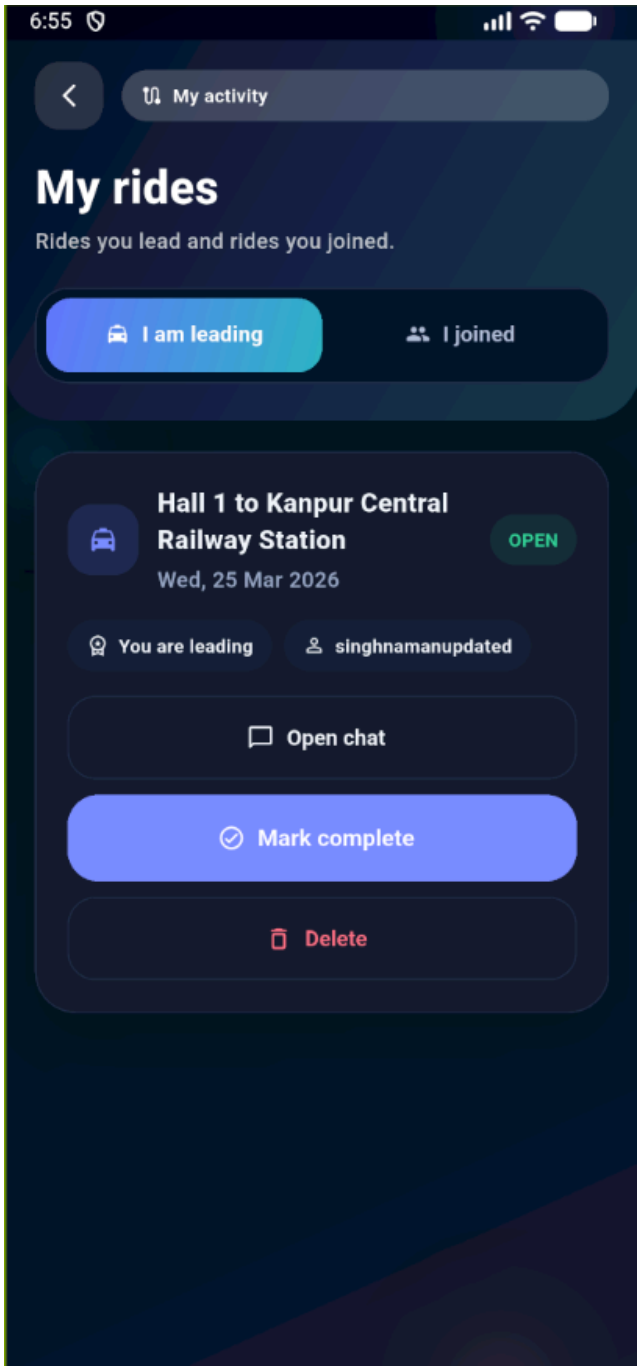
Important Notes

- All three fields (Source, Destination, Date) are mandatory. Attempting to post an incomplete form will show a red error message.
- Your display name as set in your Profile will be shown to other users on the ride card.
- You can manage, complete, or delete your posted ride from the My Rides screen at any time.

MY RIDES & RIDE DETAILS

My Rides Screen

The My Rides screen gives each user a personalised view of their ride activity. It is divided into two tabs and provides all management actions available to ride leaders and joined participants.



Tabs

- **I am leading — Displays all rides that the current user has created. This tab shows every ride regardless of status (Open or Completed).**
- **I joined — Displays all rides where the current user has sent at least one message (i.e., they are in the participants list). This makes it easy to track rides joined through the Find a Ride screen.**

Ride Cards

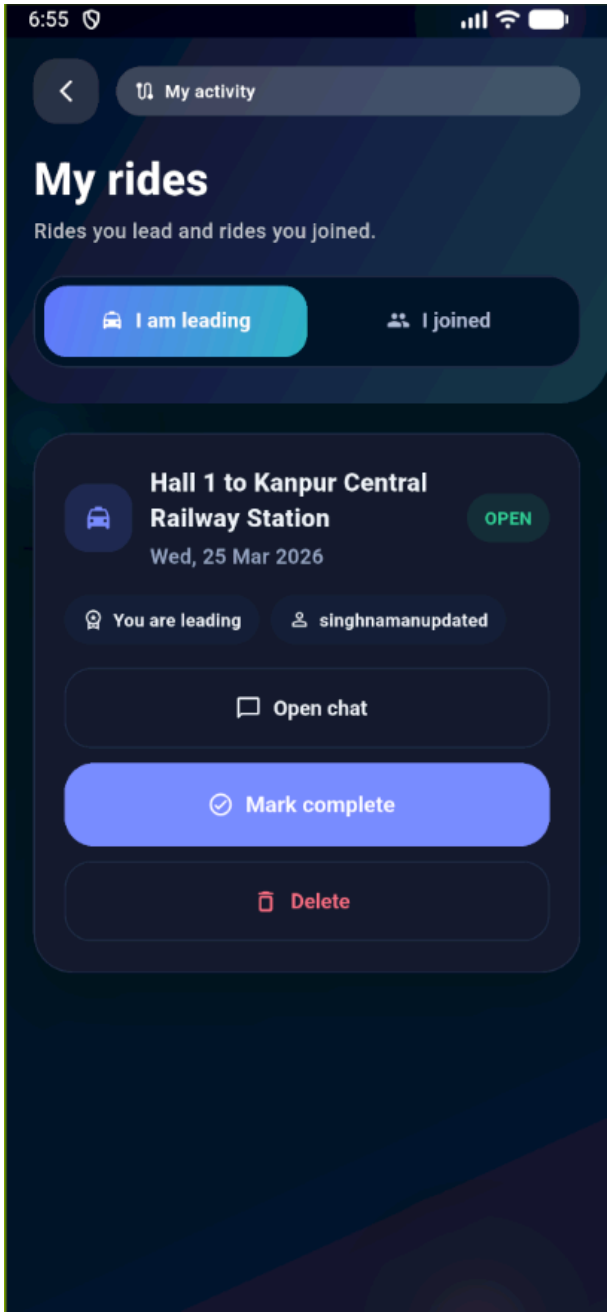
Each ride card in the My Rides screen displays:

- Route — The full source-to-destination label.
- Date — The scheduled travel date.
- Status Badge — Either 'OPEN' (green) or 'COMPLETED' (teal).
- Role Pill — Indicates whether you are the leader or a joined participant.
- Leader Name Pill — Shows the ride leader's display name.

Leader Actions

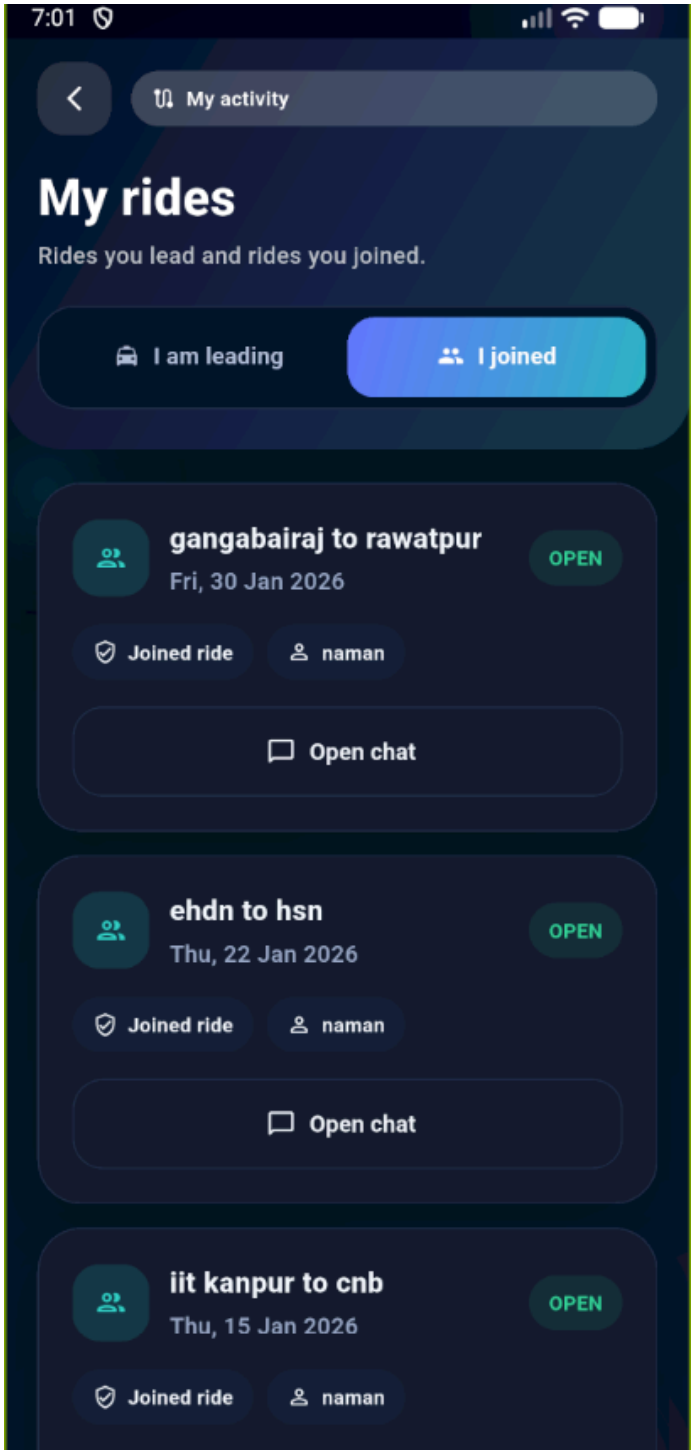
For rides displayed in the 'I am leading' tab, three action buttons are available:

- Open chat — Opens the group chat for this ride, regardless of status.
- Mark complete — Available only for open rides. Tapping this button changes the ride status to 'completed' and increments your 'Rides Completed' counter in your profile. This action cannot be undone.
- Delete — Permanently removes the ride and its data for all participants. A confirmation dialogue is shown before deletion to prevent accidental removal.



Participant Actions

For rides in the 'I joined' tab, only the 'Open chat' button is available, as participants cannot modify rides they did not create.



Delete Confirmation Dialogue

When a leader taps 'Delete' on a ride, a system dialogue box appears with the message: 'This will cancel the ride and remove it for everyone. This action cannot be undone.' The user must confirm by tapping 'Delete' or dismiss the action by tapping 'Keep'. This safeguard prevents accidental data loss.

MESSAGING IN CHATS

Ride Chat Interface

The Ride Chat screen provides a real-time, focused group messaging thread for each individual ride. It is the primary coordination tool within UniPool, allowing the ride leader and joined participants to confirm pickup points, share timings, and communicate any last-minute changes — all without needing to leave the app or create a new group elsewhere.

Key Features

- **Real-Time Messaging —** Messages are stored in and streamed from Cloud Firestore. New messages from any participant appear instantly for all members without requiring a manual refresh.
- **Message Bubbles —** Your messages appear on the right side of the screen with a blue gradient background. Messages from other participants appear on the left with a white background. Each bubble displays the sender's name and the time the message was sent.
- **Auto-Participant Registration —** The first time a user sends a message in a ride's chat, their user ID is automatically added to the ride's participants list. This is how the 'I joined' tab in My Rides is populated.
- **Multi-line Input —** The message input field expands to accommodate up to four lines, making it easy to type longer coordination messages without the input becoming cluttered.
- **Empty State —** If no messages have been sent yet, a friendly prompt is shown: 'Start the conversation with pickup details or a quick hello.'

Sending a Message

15. Tap the message input field at the bottom of the screen.
16. Type your message — pickup gate, estimated departure time, contact number, etc.
17. Tap the blue send button (arrow icon) or press the Return key on your keyboard to send.
18. Your message appears immediately as a blue bubble on the right side of the screen.

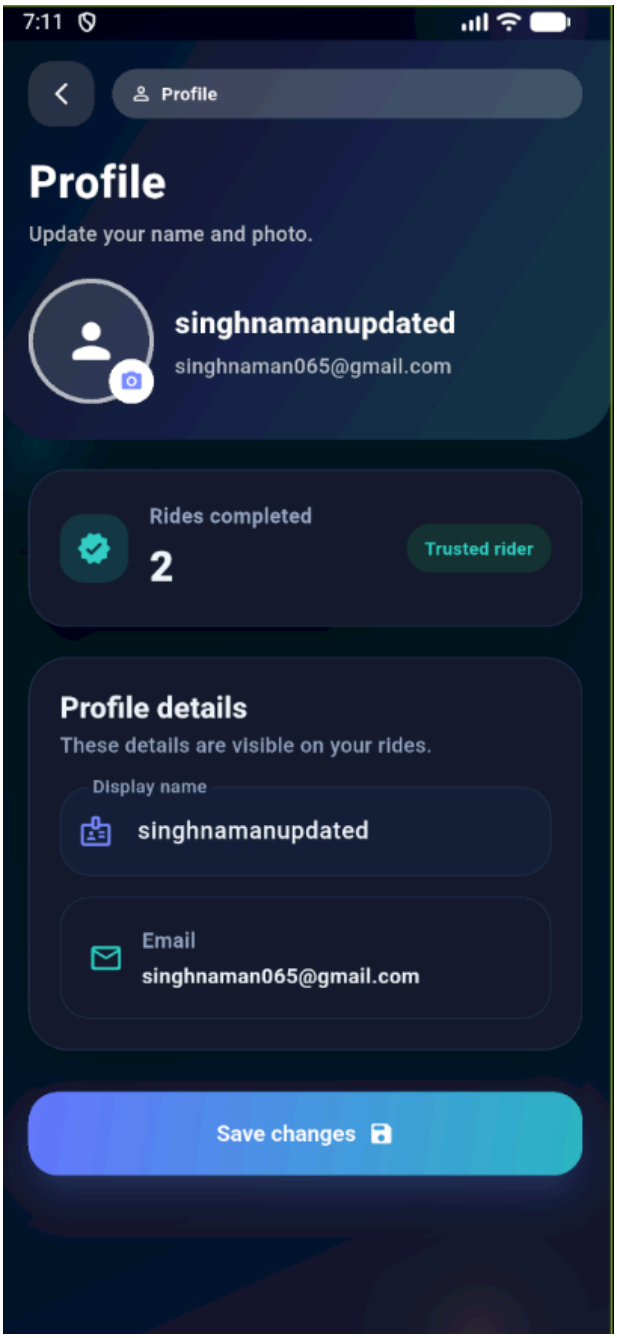
Navigation

To return from the Chat screen to the previous screen, tap the back arrow button in the top-left of the header. The chat history is preserved and can be re-opened at any time from My Rides or from the Ride Details sheet in Find a Ride.

EDIT PROFILE

Edit Profile Page

The Edit Profile page allows every UniPool user to manage their personal details and public identity on the platform. An accurate and up-to-date profile ensures that other riders can recognise and trust you.



Updating Your Profile Photo

- Tap anywhere on your circular avatar in the header area.
- Your device's image gallery will open. Select the photo you wish to use as your profile picture.
- The selected image is immediately uploaded to Firebase Storage. A loading spinner is overlaid on the avatar during the upload.
- Once the upload is complete, your new photo appears in the avatar and a green 'Profile photo updated.' confirmation is shown.

Updating Your Display Name

Your display name is the name visible to other riders on ride cards and in group chats. To update it:

- Tap the 'Display name' text field in the Profile details card.
- Clear the existing name and type your preferred name.
- Tap 'Save changes' to commit the update to the database.

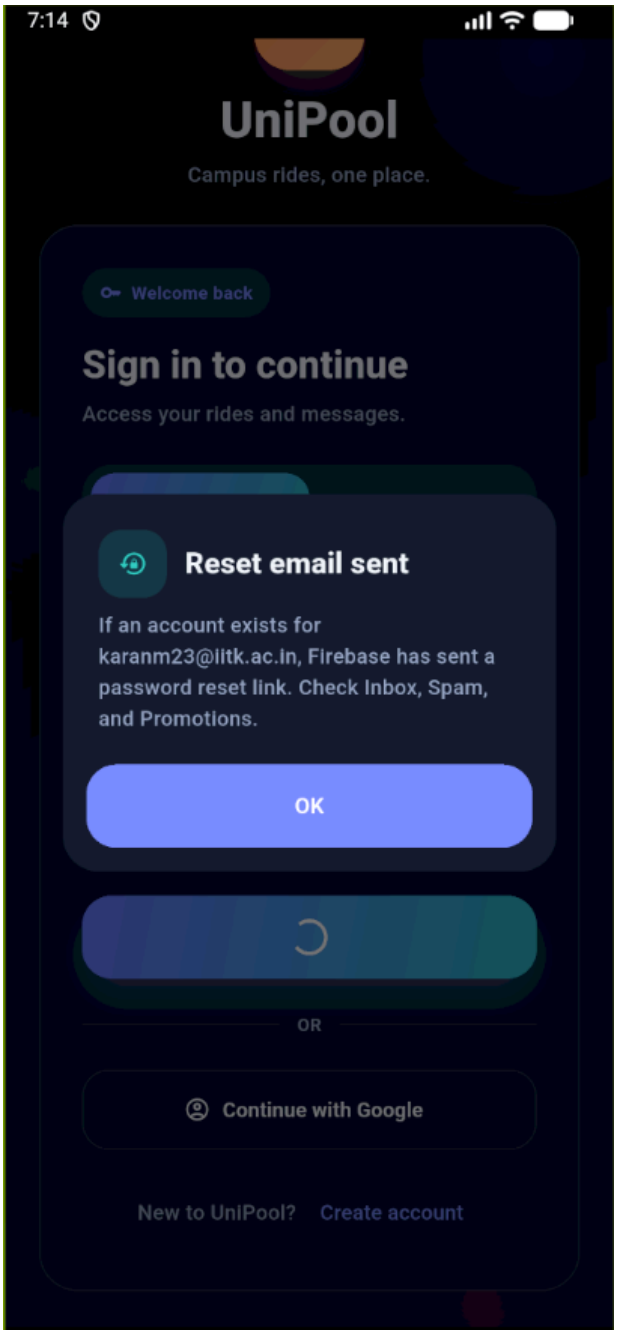
Rides Completed Counter

The 'Rides Completed' counter at the top of the Profile page is a read-only field. It is automatically incremented each time you mark one of your rides as completed from the My Rides screen. This counter is also visible to other users when they view your ride details in Find a Ride, helping establish your reputation as a reliable ride leader.

CHANGE / FORGOT PASSWORD

Password Reset (Forgot Password)

If you have forgotten your password and cannot sign in, UniPool provides a secure, email-based password reset flow powered by Firebase Authentication.



Steps to Reset Your Password

19. On the Welcome Page, ensure the 'Sign in' tab is selected.
20. Enter your registered college email address in the 'College email' field.
21. Tap the 'Forgot password?' text button below the password field.
22. A password reset link will be sent to your email address. A confirmation snackbar reading 'Password reset link sent. Check your inbox.' will appear.
23. Open the reset email and click the secure link. You will be taken to a Firebase-hosted page where you can set a new password.
24. Return to the UniPool app and sign in with your new password.

Important Notes

- The reset email is only sent if the address entered matches a registered account.
- The reset link expires after a short period for security reasons. If it has expired, simply repeat the process to receive a new link.
- If your email field is empty when you tap 'Forgot password?', an error message will prompt you to enter your email first.

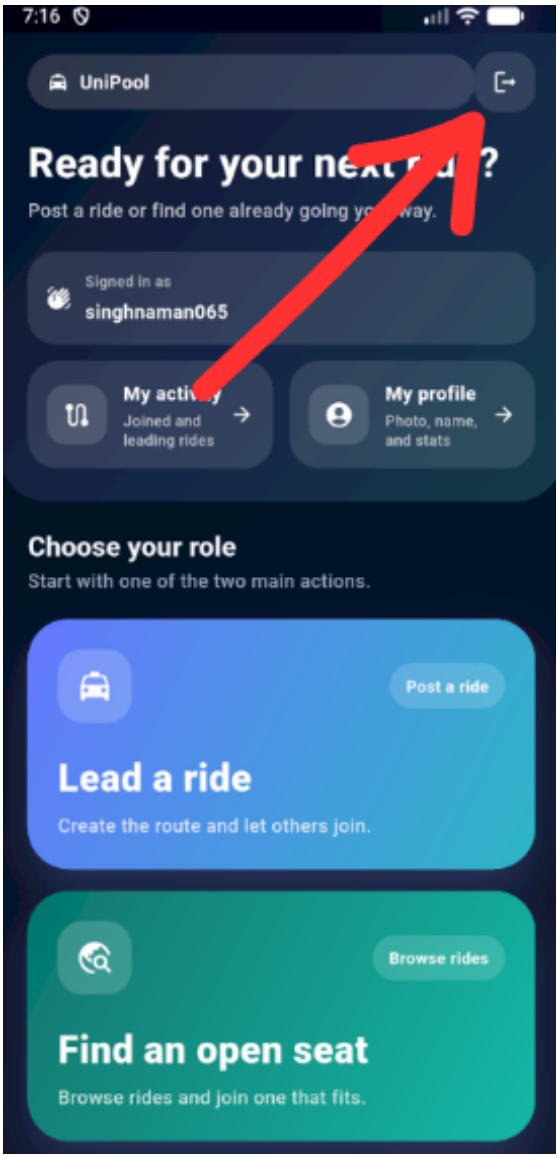
Change Password

To change your password while you are already logged in (for example, as a routine security update), you must currently sign out, use the 'Forgot Password' flow described above, and then sign back in with the new credentials. A dedicated in-app password change screen is planned for a future release.

LOGOUT CONFIRMATION

Logout

UniPool provides a simple and immediate logout mechanism accessible from the Home Screen. The action is designed to be direct and instantaneous, since all ride data is stored in the cloud and is not affected by signing out.



How to Log Out

25. From the Home Screen, locate the logout icon (arrow-out-of-box icon) in the top-right corner of the header.

26. Tap the icon. UniPool will immediately sign you out via Firebase Authentication.
27. The app will navigate back to the Welcome Page, ready for a new sign-in.

What Happens After Logout

- Your ride data, chat history, and profile information are all safely stored in Firebase and will be fully restored on your next login.
- No local data is retained on the device after logout.
- Any other users currently viewing a ride you created will still see your ride and can continue chatting; your logout does not affect others.

Session Persistence

UniPool uses Firebase's built-in session persistence. If you close the app without logging out, you will be automatically returned to the Home Screen the next time you open the app, without needing to sign in again. The app displays a brief loading animation while it verifies your session state.